Lösung: Linearfaktorzerlegung eines komplexen Polynoms

$$f(z) = 2z^4 - 2z^3 - 2z^2 - 2z - 4$$

1. Nullstelle erraten: $z_0 = -1$

$$\frac{f(z)}{z - z_0} = 2z^4 - 2z^3 - 2z^2 - 2z - 4 : (z+1) = 2z^3 - 4z^2 + 2z - 4$$

$$\begin{array}{r} -(2z^4 + 2z^3) \\ \hline -4z^3 - 2z^2 - 2z - 4 \\ -(-4z^3 - 4z^2) \\ \hline 2z^2 - 2z - 4 \\ -(2z^2 + 2z) \\ \hline -4z - 4 \\ -(-4z - 4) \\ \hline 0 \end{array}$$

2. Nullstelle erraten: $z_1 = 2$

$$2z^3 - 4z^2 + 2z - 4 : (z-2) = 2z^2 + 2 = 2(z^2+1)$$

$$\begin{array}{r} -(2z^3 - 4z^2) \\ \hline 2z - 4 \\ -(2z - 4) \\ \hline 0 \end{array}$$

$$= 2(z-i)(z+i)$$

$\Rightarrow$ Linearfaktorzerlegung:

$$f(z) = (z+1)(z-2)(z+i)(z-i)$$